Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design) Course Overview

Spring 2026

Instructor: Hao Li

Words Before The Journey

- Teaching Language (speaking)
  - ~~Completely in Chinese, or mainly in Chinese~~
  - Mainly in English, yet partially in Chinese for assistance?
  - Completely in English?
- Code In Class 外弛内张、外圆内方
  - Be at ease physically
    - Feel free to interrupt for posing related questions, initiating communications, etc. --- Just note that you had better put up your hand first (to catch my attention) and then continue
    - Feel free to go to the WC during class, but just keep your action as quiet as possible
    - Feel free to eat or drink (if not forbidden by SBI) in class, but also just keep quiet
  - Be serious spiritually
    - Study conscientiously
    - Respect course arrangements

Words Before The Journey

- Office Hours
  - Mon & Wed 14:00-16:00, Tue 9:00-11:00 @ Lixi 201-4
- Communication
  - Face-to-face
    - During office hours
    - During pre-, inter-, post-class times
  - By email: haoli@ahu.edu.cn
    - When writing the email title, you had better write it as “ISE 333 UI 2026: {Title content}” to better catch my attention, namely with the prefix “ISE 333 UI 2026” added
    - If do not reply to your email (perhaps because I have not noticed your email), feel free to remind me face-to-face at proper times
  - Teams code: e213lg1
  - ~~WeChat~~

2025-2026-2 Global Faculty Office Hour

Fang Li	理西212	周三11:30-13:30	周四11:30-13:30	周五11:30-13:30
UNGTAEK LIM	理西201-5	周二14:00-15:40	周三14:00-15:40	周五14:00-15:00
fqat Ur Rehman	理西201-5	周二14:00-17:00	周四10:00-13:00	
ing Tiew On	理西201-5	周三9:00-12:00	周三14:00-17:00	
ieh Avazzadeh	理西404	周二14:00-18:00	周五14:00-18:00	
贺从笑	理西212	周四14:00-16:00	周五8:00-10:00	
李颢	理西201-4	周一14:00-16:00	周二9:00-11:00	周三14:00-16:00
于心瑞	理西201-4	周一9:30-11:00	周一11:30-12:30	周二11:30-12:30

Course Overview

- Course Description
  - This course provides an introductory knowledge of user interface (UI) design
    - Indispensable in achievement of a large variety of complete software and hardware systems in practical applications
  - This course covers a comprehensive list of fundamental topics
    - Usability of interactive systems and universal usability
    - Guidelines, principles, and theories
    - Design, evaluation, and the user experience
    - Direct manipulation, immersive environments, and fluid navigation
    - Expressive human and command languages
    - Devices, communication, and collaboration
    - Documentation, user support, information search, and data visualization

Course Overview

- Course Description
  - This course provides an introductory knowledge of user interface (UI) design
  - This course covers a comprehensive list of fundamental topics
  - This course tends to arouse your systematic and application-oriented thinking
    - In the light of the instructor's own academic and industrial experiences, this course also adds pertinent knowledge of industrial-level flavour
    - Facilitating development and enhancement of your ability to flexibly apply learned knowledge of UI design.
- Textbook
  - Ben Shneiderman, "*Designing the User Interface: Strategies for Effective Human-Computer Interaction*" (6th Edition), 2018
- Teaching Style
  - Textbook-Based & Personally-Interpreted/Enriched (TBPIE or "Too Big PIE")

Some knowledge points presented in the teaching PPTs are not given in the textbook

Course Overview

- Course Objectives
  - Get acquainted with fundamental concepts in the context of UI design and use them proficiently for potential academic and industrial communications
  - Have a deep consciousness of user-centered and practical application-oriented thinking for UI design and have a deep consciousness of applying the learned knowledge flexibly in practice
  - Know existing representative guidelines, principles, and theories of UI design and know systematic method of UI evaluation
  - Provide appropriate ways of documentation, information search, and data visualization in the context of UI design

Course Overview

- Course Schedule

Week	Topic	
Week 2	• Chap.1: Usability of interactive systems	
Week 3	• Chap.2: Universal usability	
Week 4	• Chap.3: Guidelines, principles, and theories	
Week 5	• Chap.4: Design + **Quiz 1**	
Week 6	• Chap.5: Evaluation and the user experience	
Week 7	• Chap.7: Direct manipulation and immersive environments	
Week 8	• Chap.8: Fluid navigation	
Week 9	• Chap.9: Expressive human and command languages	• **Mid Exam**
Week 10	• Chap.10: Devices	
Week 11	• Chap.11: Communication and collaboration	
Week 12	• Chap.12: Advancing the user experience + **Group Homework Out**	
Week 13	• Chap.14: Documentation and user support + **Quiz 2**	
Week 14	• Chap.15: Information search	
Week 15	• Chap.16: Data visualization	
Week 16	• Course review + **Group Homework Submission**	
Week 17	• **Group Homework Evaluation**	• **Final Exam**

Course Overview

- Course Evaluation
  - Memory-style examination points are totally based on teaching PPTs 硬知识
    - Each PPT is released after its class (for sake of updating with contingent class feedback)
    - For concepts and terms that I mention when giving examples or narrating some science-technology stories, if they do not appear in PPTs, they will not be examined --- no worry
  - Knowledge-application-style examination points can be flexible 软知识
    - New application cases and scenarios not presented in teaching PTTs can appear in exams, only if the knowledge to handle them is learned and presented in teaching PPTs
  - Group homework pre-info
    - 5 students / group (you may prepare to form your group from now on)
    - For remaining students who have not joined any group by certain date, I will assign them randomly either to form new groups or to existing groups, depending on their number
  - Homework submission
    - Paper version (by printing or handwriting) with your names signed
    - Submitted in class during the roll call on the deadline date

Course Overview

- Course Evaluation
  - Memory-style examination points are totally based on teaching PPTs
  - Knowledge-application-style examination points can be flexible
  - Group homework pre-info
  - Homework submission

Composition of Grade	Percentage	Dates
Quiz 1	10	• Week 5
Quiz 2	10	• Week 13
Group Homework (5 people/group)	20	• Week 12-17
Mid Exam	30	• Week 9
Final Exam	30	• Week 17

- Extra Words
  - Academic integrity
  - PPT sytle
    - Teaching PPTs are intentionally enriched with words and even made “verbally sollen”, out of the consideratiion that they serve as the unique baseline for exams and they had better be so to facilitate your review of learned knowledge
    - But be aware that such style is by no means always appropriate, especially in research and industry

DISCUSSION

Q & A

ISE 333: User Interface Development (Design)
Chapter 1: Usability of Interactive Systems

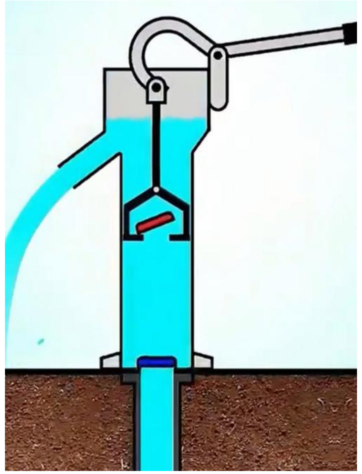
Spring 2026

Instructor: Hao Li

Introduction

Have you realized a very good example of user interface just around us nearby?

- Conception
  - What is the user interface?
    - The user interface (or UI for short) refers to the medium/media via which users interact with computer systems (or other kinds of systems such mechanical systems, traffic systems, administrative systems, cultural systems etc.), or in other words, via which the systems provide services to the users. It encompasses all the elements involved in the interaction between the users and the systems
    - Our focus is computer systems, but knowledge essence is more generally applicable



Pull-press well

Tri-branch handrail

Infrastructure

Administration hall

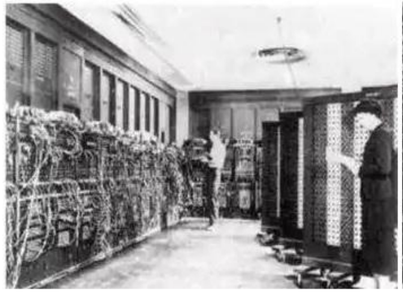
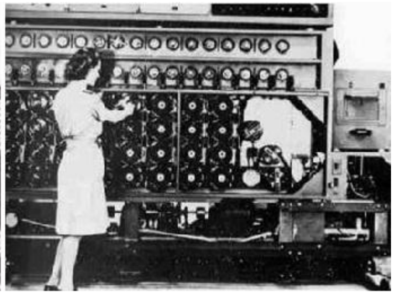
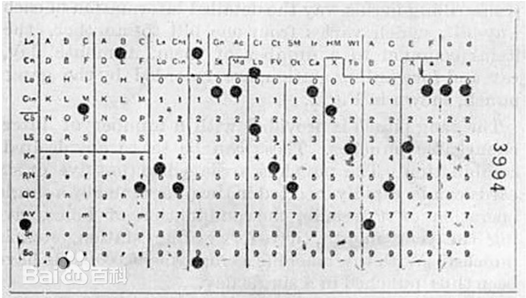
Il y a un spectacle
plus grand que la mer,
c'est le ciel;
il y a un spectacle
plus grand que le ciel,
c'est l'intéreiur de l'âme.

--- Les Misérables,
by Victor Hugo

Language

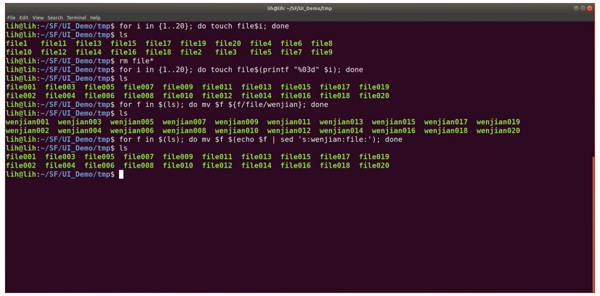
Introduction

- Revolution and Contribution
  - User-interface designers are the heros of a profound transformation
    - (From a historical perspective) Their work turned personal computers into today's wildly successful mobile devices, enabling users to communicate and collaborate in remarkable ways. The desktop applications that once served the needs of professionals have increasingly given way to powerful social tools that deliver compelling user experiences to global communities. These invigorated communities conduct business, communicate with family, get medical advice, and create user-generated content that can be shared with billions of connected users.
  - Are their contribution (or in other words, is the contribution of user interface design) really so remarkable?
    - To deeply comprehend above "tribute to UI design", let's take a time machine to go back

Early computer
& puch card

Demo: "magic" of CLI
(command line interface)



Introduction

Core technology had not changed,
but what made a huge difference is the UI

- Revolution and Contribution
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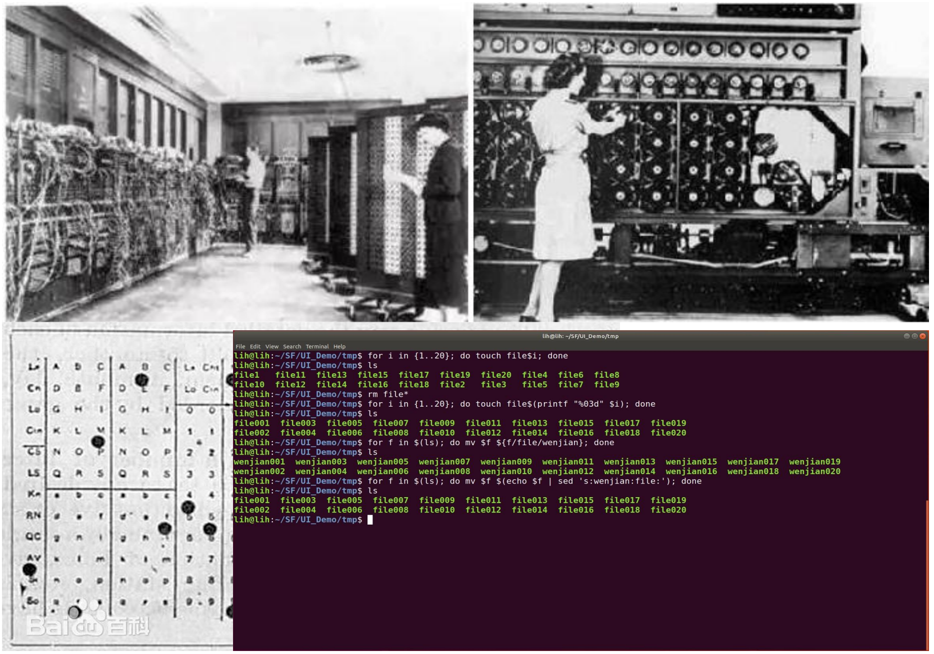
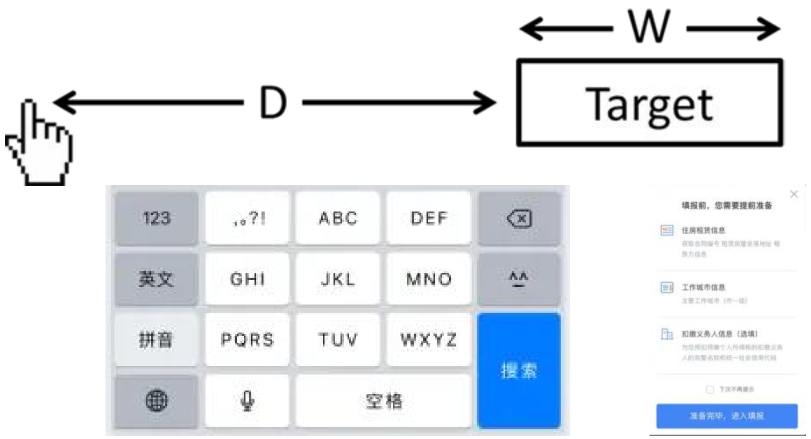
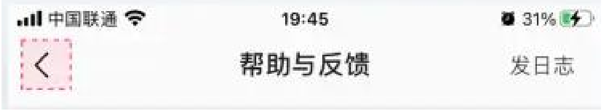


Introduction

- Revolution and Contribution
  - Life-changing shifts were made possible because researchers and user-interface designers (developers) harnessed technology to serve human needs
    - Researchers created the interdisciplinary science of *human-computer interaction* (HCI) by applying the methods of experimental psychology to the powerful tools of computer science. Then they integrated lessons from educational and industrial psychologists, instructional and graphic designers, technical writers, experts in human factors or ergonomics, and growing teams of anthropologists and sociologists.

朝发轫于苍梧兮
夕余至乎县圃
欲少留此灵琐兮
日忽忽其将暮
吾令羲和弭节兮
望崦嵫而勿迫
路漫漫其修远兮
吾将上下而求索





Miller's law (7±2 rule)

Fitts's law : T = a + b * log (D/W + 1)

Sequence effect

Introduction

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Tri-branch handrail

Keyboard

Il y a un spectacle
plus grand que la mer,
c'est le ciel;
il y a un spectacle
plus grand que le ciel,
c'est l'intéreiur de l'âme.

--- Les Misérables,
by Victor Hugo

Huffman coding

a : 1101	c : 1110
d : 10010	e : 00
g : 111100	i : 0111
l : 101	m : 100110
n : 0110	p : 11111
q : 111101	r : 0100
s : 1000	t : 0101
u : 1100	y : 100111

Introduction

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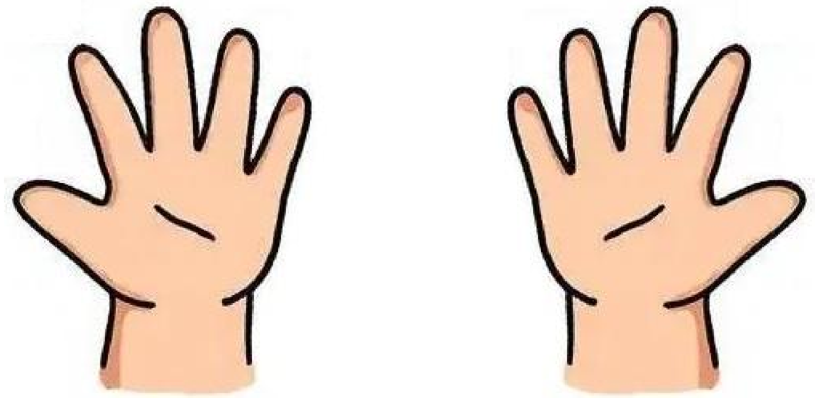
Distance



Left hand vs. right hand

Introduction

- Social Impact
  - At an individual level, effective user experiences change people's lives
    - Doctors diagnoses
    - Pilots flying airplanes
    - Students learning knowledge
    - Teachers imparting knowledge

Health code in Zhi Fu Bao

  - At a societal level, connected communities open up new forms of collective action and policy engagement
    - Having more informed citizens may lead to better decisions, more transparent governance, and greater equity when facing legal, health, or civic challenges
  - The steadily growing interest in human-computer interaction stems from the designers' desire to improve the users' experience
- User Interface Development/Design (UID) is indispensable

Introduction

- UID In Information Era
  - The remarkably rapid and widespread adoption of mobile devices (including smartphones, tablets, game devices, fitness trackers, etc.) supports personal communication, collaboration, and content creation
  - What is happening in the realms of social networking and user-generated content is experiencing explosive growth
    - Networked media (e.g. Bilibili, Youtube, Dou Yin), networked commerce (e.g. Jing Dong, Tao Bao, Pin Duo Duo), networked communication (e.g. Teams, WeChat)
  - Sociologists, anthropologists, policymakers, and managers are attentive
    - They are studying how social media are changing education, family life, shopping, and services such as medical care, financial advice, and political organizaitons
    - They are dealing with issues of organizational impact, job redesign, distributed teamwork, work-at-home scenarios, and long-term societal changes
    - As face-to-face interaction gives way to screen-to-screen, how can personal trust be preserved? How can empathy be conveyed? How can civic participation be enhanced?

Introduction

- UID In Information Era
  - UID faces the challenge of providing services on small-, wall-, and mall-sized displays, ranging widely across various products for various people of various cultures and of various personalities
  - Plasticity and malleability have to be respected ... hmm ... if universally impossible, but at least as much as possible
    - Each individual has their own “thinking singularity”
  - Old forms of UI are disappearing, whereas new forms are appearing quickly
    - New user interfaces become ubiquitous, pervasive, invisible, and embedded in the surrounding environment
    - They can be context-aware, attentive, and perceptive, sensing users’ needs and providing feedback through ambient displays
    - They can also be wearable, control implanted (under-the-skin), multi-modal, gestural, and affective (responding to the user’s emotional state)

Introduction

Are outcomes of UID always desirable?

- Embrace UID
  - The inspirational pronouncements from technology prophets can be thrilling, but rapid progress is more likely to come from those who do the hard work of tuning UID to genuine human needs
  - The next phase of human-computer (other systems) interaction will be strongly influenced by those who are devoted to broadening the community of users by promoting universal usability and facilitating many forms of social media and network participation
  - User interfaces that deliver excellent user experiences will be a key component in improving healthcare, creating sustainable economies, protecting natural resources, and resolving conflicts

Usability Goals and Measures

- Usability Goals
  - Usability
  - Universality
  - Usefulness
    - They represent primary features of high-quality UID
- Achievement of Usability Goals
  - Thoughtful planning
  - Sensitivity to user needs
  - Devotion to requirements analysis
  - Diligent testing
    - Within budget and on schedule

Dialectic (positive<->negative) thinking

- Inverse thinking
- Displaced, position-changing thinking (perspective taking)
- Essence-grasping thinking
- ... ...

Usability Goals and Measures

- Roles of Managers

Enterprise/industrial interfaces

  - Managers who pursue user-interface excellence first select experienced designers and then prepare realistic schedules that include time for requirements gathering, guidelines preparation, and repeated testing
- Roles of Designers
  - Designers begin by determining user needs, generating multiple design alternatives, and conducting extensive evaluations
    - Modern user-interface-building tools enable implementations to quickly build working systems for further testing
  - Successful designers have a thorough understanding of the diverse community of users and the tasks that must be accomplished
    - They go beyond vague notions of “user friendliness”, “intuitive”, and “natural”, doing more than simply making checklists of subjective guidelines

“go beyond” but not “repel”

Usability Goals and Measures

- Roles of Managers
- Roles of Designers
  - Designers begin by determining user needs, generating multiple design alternatives, and conducting extensive evaluations
  - Successful designers have a thorough understanding of the diverse community of users and the tasks that must be accomplished
  - They study evidence-based guidelines and pursue the research literature when necessary
  - They are also aware of the importance of eliciting emotional responses
- Ideal
  - In the best cases, the UI almost disappears, enabling users to concentrate on their work, exploration, or pleasure

大道至简

Usability Goals and Measures

- Usability Measures
  - Close interaction with the user community leads to a well-chosen set of benchmark tasks that is the basis for usability goals and measures. For each user type and each task, precise measurable objectives guide the designer through the testing process
    - ISO 9241 standard *Ergonomics of Human-System Interaction* (ISO, 2013)
  - The ISO 9241 focuses on admirable goals
    - Effectiveness
    - Efficiency
    - Satisfaction
  - Following usability measures, which focus on the latter two goals, lead more directly to practical evaluation

Usability Goals and Measures

- Usability Measures
  - Time to learn
    - How long does it take for typical members of the user community to learn how to use the actions relevant to a set of tasks?
  - Performance
    - Speed: How long does it take users to carry out the benchmark tasks?
    - Quality: How well does it take users to carry out the benchmark tasks?
  - Rate of errors by users
    - How many and what kinds of errors do users make in carrying out the benchmark tasks?
    - Error handling by users: Although time to make and correct errors might be incorporated into the speed of performance, error handling is such a critical component of UI usage that it deserves extensive study
    - Attention: not confused with handling of UID bugs by designers/developers
    - Impact of error handling

Usability Goals and Measures

- Usability Measures
  - Time to learn
  - Performance (especially speed of performance)
  - Rate of errors by users
  - Retention over time
    - How well do users maintain their knowledge after an hour, a day, or a week?
    - Retention over time may be linked closely to time to learn, and frequency of use plays an important role as well
  - Subjective satisfaction
    - How much did users like using various aspects of the UI?
    - The answer can be ascertained by interviews or by written surveys that include satisfaction scales and space for free-form comments

Usability Goals and Measures

- Usability Measures
  - Time to learn
  - Performance (especially speed of performance)
  - Rate of errors by users
  - Retention over time
  - Subjective satisfaction
- Forced tradeoffs
  - Difficult and even impossible to succeed in every measure
    - If lengthy learning is permitted, task performance tends to be improved
    - If the rate of errors is to be kept extremely low, speed of performance tends to be sacrificed
  - Managers and designers who are aware of the tradeoffs can be more effective if they make their choices explicit and public, e.g. by requirements documents

Usability Goals and Measures

- Usability Measures
- Forced tradeoffs
- Rules of Thumb
  - A thoroughly documented set of user needs clarifies the design process
  - A carefully tested prototypes generates fewer changes during implementation while avoiding costly updates after release
  - Thorough acceptance testing of the implementation produces robust UIs that are aligned with user needs
  - Continuous evaluation based on usage logs and user comments guide evolutionary refinements

Usability Goals and Measures

- Usability Measures
  - Time to learn
  - Performance (especially speed of performance)
  - Rate of errors by users
  - Retention over time
  - Subjective satisfaction
- Discussion
  - Analyse usability measures of the UNIX/Linux command line interface (CLI)
  - Analyse usability measures of Chinese, English, etc. and perform a comparative study

DISCUSSION

Q & A

Usability Motivations

- Interest in UI usability 见贤思齐
  - The enormous interest in UI usability arises from the demonstration of the benefits that come from well-designed UIs
- Consumer Electronics, E-Commerce, and Social Media
- Games and Entertainment
- Professional Environments
- Exploratory, Creative, and Collaborative Interfaces
- Sociotechnical Systems

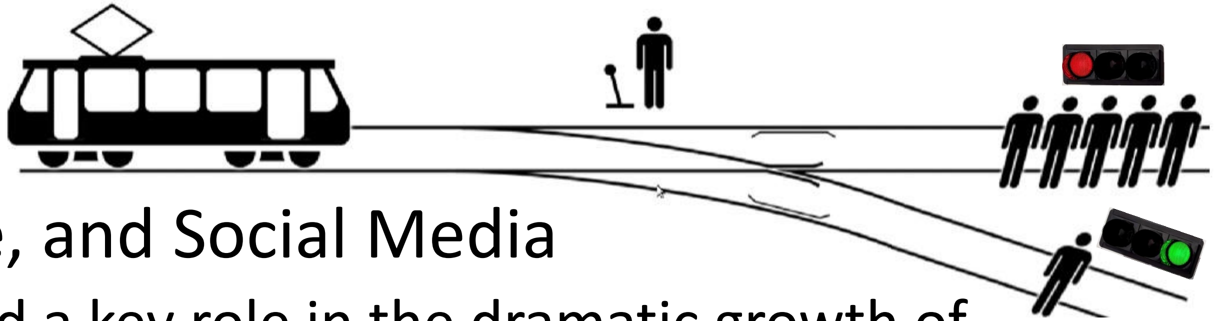
Usability Motivations

- Consumer Electronics, E-Commerce, and Social Media
  - User experience designers have played a key role in the dramatic growth of consumer electronics by providing effective and satisfying designs that have become widely adopted for personal communications, education, healthcare, and much more
  - User-generated content such as online restaurant, film, or product reviews have become part of daily life for many users
  - Critics raise concerns about reduced privacy, dangers in distracted driving, and declining quality of interpersonal relationships



Usability Motivations

The Trolley Problem (1905)



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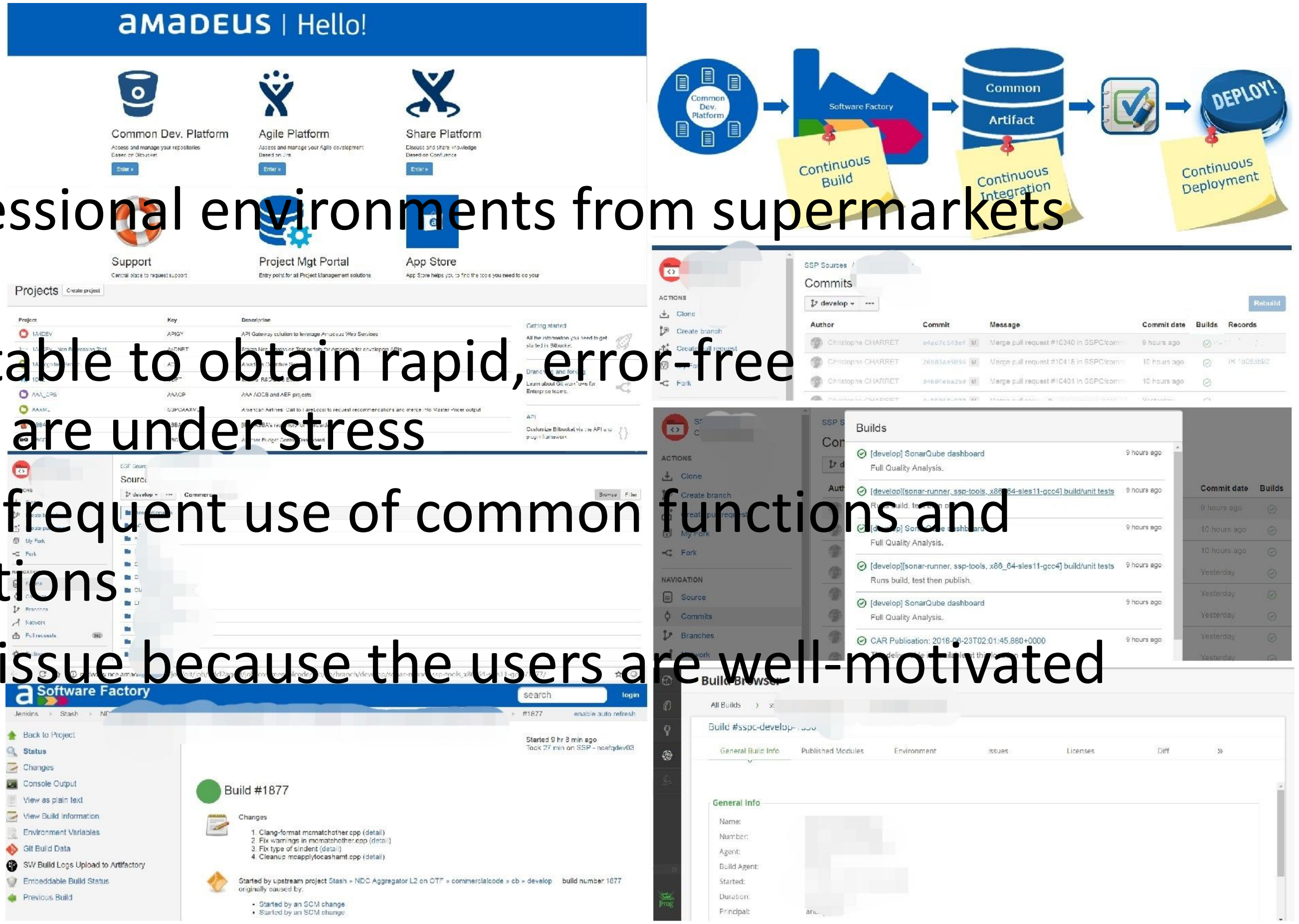
  - User-generated content such as online restaurant, film, or product reviews have become part of daily life for many users
  - Critics raise concerns about reduced privacy, dangers in distracted driving, and declining quality of interpersonal relationships
    - At a societal level, it is difficult and unfair to oblige UI designers to predict and even be responsible for undesirable or bad outcomes of their UIs
    - At an individual level, UI designers should have a deep conciousness of social responsibility and try their best to avoid undesirable or bad outcomes of their UIs

Individual survival <- vs -> Social harm

Individual dilemma -> jump out -> Organization/Government/Country/Society-level resolution

Usability Motivations

- Games and Entertainment
  - The rapid expansion of home and entertainment applications is a further source of interest in usability
- Professional Environments
  - Users also benefit from UIs in professional environments from supermarkets to space stations
  - Lengthy training periods are acceptable to obtain rapid, error-free performance, even when the users are under stress
  - Retention over time is obtained by frequent use of common functions and practice sessions for emergency actions
  - Subjective satisfaction is less of an issue because the users are well-motivated professionals



Usability Motivations

- Exploratory, Creative, and Collaborative Interfaces
  - An increasing fraction of computer use is dedicated to supporting open-ended exploration that promotes human creativity while lowering barriers to collaboration
  - Exploratory applications include web browsers, search engines, data visualization, and team collaboration support
  - Creative applications include design environments, music-composition tools, animation builders, and video-editing systems
  - Collaborative interfaces include online face-to-face meetings, webinars (web seminars), on-cloud sharing tools
- Sociotechnical Systems
  - UIs for these systems, often created by governmental organizations, have to deal with trust, privacy, responsibility, and limiting tampering, deception etc.

Why created by governmental organizations?

Goals for UI Developers/Designers’ Profession

- Influencing Academic and Business Researchers
  - Researchers in business, management, education, sociology, anthropology, and other disciplines are benefiting from and contributing to the study of human-computer interaction
  - Fruitful directions for research
    - Reduced anxiety and fear of computer usage
    - Graceful evolution (from novice to knowledgeable user to expert)
    - Social media
    - Input devices
    - Information exploration

Goals for UI Developers/Designers’ Profession

- Influencing Academic and Business Researchers
- Providing Tools-Techniques-Knowledge (TTK) for UIDers
  - User-interface development and design (UID^2) are hot topics, and international competition is lively.
  - There is a great thirst for knowledge about software tools, design guidelines, and testing techniques

PK

Goals for UI Developers/Designers' Profession

- Influencing Academic and Business Researchers
- Providing Tools-Techniques-Knowledge (TTK) for UIDers
- Raising the General Public's UI Consciousness
  - Encourage users to translate their internal fears into outraged action别内耗
    - Instead of feeling guilty when they get an error message, users should express their anger at the UIDer who was so inconsiderate and thoughtless
    - Instead of feeling inadequate or foolish because they cannot remember a complex sequence of actions, users should complain to the UIDer who did not provide a more convenient mechanism or should seek another product that does

Usability of Interactive Systems

- Review
  - Basics, usability goals, usability measures, usability motivations
- Practice
  - Among the following goals, which is not a usability goal?
    - [1] Universality; [2] Uniqueness; [3] Usability; [4] Usefulness
  - The terms “to B” (to business), “to C” (to customers), and “to G” (to government) have already stepped into people’s daily-life conception. Then
    - The usability motivation of consumer electronics, e-commerce, and social media is usually what of “to B/C/G”?
    - Professional environments / sociotechnical systems are what of “to B/C/G” respectively?
    - Exploratory, creative, and collaborative interfaces can be what of “to B/C/G”?
  - For UIs dedicated to profession environments, which of the usability measures are normally cared more than the others?

DISCUSSION

Q & A